VII SEMESTER

DEPARTMENT OF ECE
COLLEGE OF ENGINEERING GUINDY CAMPUS, ANNA UNIVERSITY, CHENNAI
EC23702– Mini Project
PROJECT PROPOSAL

TEAM NO.:

Title of the Project:	**Lightweight AUTOSAR-Inspired Layered Firmware Architecture for Distributed Embedded Systems**		
Project Type:	**Internal**		
Name of the Project Members:	**Member 1 (M1)** **Pragadeesh R**	**Member 2 (M2)** **Jagakishan S K**	**Member 3 (M3)** **Hariharan S**
Registration Number:	**2023105512**	**2023105513**	**2023105528**
Name of the Supervisor:	**Dr. K Praveen**		
Designation:	**Assistant Professor (Sl. Gr.)**		

ABSTRACT (150 – 200 words)

Embedded applications developed for microcontrollers are frequently designed using a monolithic architecture, where application logic, peripheral drivers, communication mechanisms, and system services are tightly coupled. As embedded systems evolve into distributed, multi-controller architectures, such tightly integrated designs become increasingly difficult to scale, maintain, reuse, and integrate. Although industrial frameworks such as AUTOSAR address these challenges through standardized layered software architecture, their complexity, licensing requirements, and extensive tooling make them impractical for academic learning and experimentation. This project presents the design and implementation of a lightweight, AUTOSAR-inspired firmware framework using FreeRTOS on STM32 microcontrollers. The framework recreates key AUTOSAR concepts, including layered software architecture, software component abstraction, runtime communication, driver initialization, and reusable basic software services, while remaining sufficiently lightweight for educational use. Multiple STM32-based Electronic Control Units (ECUs) communicate over a distributed network to demonstrate modular application development, hardware abstraction, and real-time task execution. The project provides an educational platform for understanding modern distributed system architecture and demonstrates how standardized layering improves software modularity, portability, and maintainability in distributed embedded systems.

Project Classification:	Experimental	**Hardware/Software requirement:**	**ARM Cortex-M Series Boards, SEGGER SystemView**

Signature of the Project Members **Signature of the Supervisor**

Approved /Not Approved

Signature of the Panel Members